LIMITS OF INVERSE SYSTEMS OF CLOSED IMMERSIONS ARE CLOSED IMMERSIONS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 985-999, following the statement at 971-983.

Editorial note. No omissions; the filtered-colimit-of-surjections observation is displayed in the source.

Allowed standard prerequisites (editorial).

- Existence of directed inverse limits. lemma-directed-inverse-system-has-limit (same chapter, card L26291, cited at ~988-989). lim X_i and lim Y_i exist.

Source reference. Original lemma 07SF: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 878–878. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let I be a directed set. Let $(X_i, f_{i'i}) \to (Y_i, g_{i'i})$ be a morphism of inverse systems over I of algebraic spaces over S. Assume*

(1) *the morphisms $f_{i'i} : X_{i'} \to X_i$ are affine,*
(2) *the morphisms $g_{i'i} : Y_{i'} \to Y_i$ are affine,*
(3) *the morphisms $X_i \to Y_i$ are closed immersions.*

Then $\lim X_i \to \lim Y_i$ is a closed immersion.

Proof. Observe that $\lim X_i$ and $\lim Y_i$ exist by Lemma 07SF. Pick $0 \in I$ and choose an affine scheme V_0 and an étale morphism $V_0 \to Y_0$. Then the morphisms $V_i = Y_i \times_{Y_0} V_0 \to U_i = X_i \times_{Y_0} V_0$ are closed immersions of affine schemes. Hence the morphism $V = Y \times_{Y_0} V_0 \to U = X \times_{Y_0} V_0$ is a closed immersion because $V = \lim V_i$, $U = \lim U_i$ and because a limit of closed immersions of affine schemes is a closed immersion: a filtered colimit of surjective ring maps is surjective. Since the étale morphisms $V \to Y$ form an étale covering of Y as we vary our choice of $V_0 \to Y_0$ we see that the lemma is true. □

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.